DISCOVER · wide-field surveys find $\mathcal{O}(1000)$ lensed AGN

CHARACTERIZE

Rubin LSST

PSF FWHM 0.71–0.77″

g r i · 10-yr

Euclid

PSF FWHM 0.16–0.5″

VIS Y J H · wide

Roman

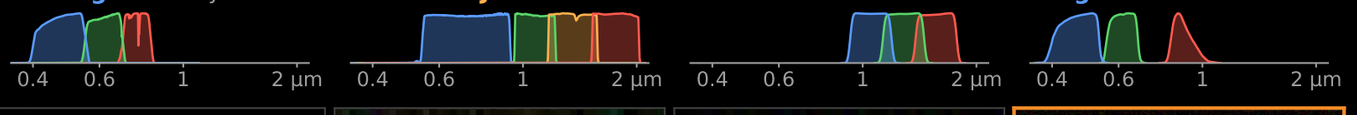
PSF FWHM 0.087–0.13″

F106 F129 F158 · HLWAS

Lazuli WCC

PSF FWHM 33–62 mas

LZg LZr LZz · 4×300 s



Quad z_d=0.89 z_s=2.74

θ_E = 0.84 arcsec

1 arcsec

IFS field

D
G
C
A
B

Quad z_d=1.16 z_s=2.64

θ_E = 0.23 arcsec

1 arcsec

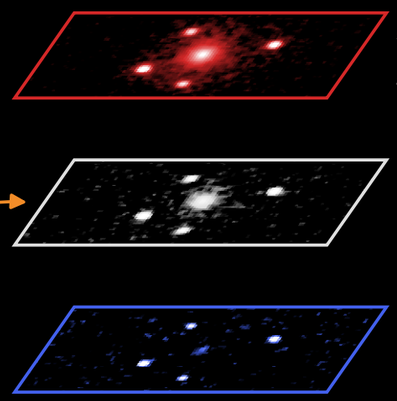
Double z_d=0.57 z_s=1.22

θ_E = 0.74 arcsec

1 arcsec

LAZULI IFS · one pointing, every image

0.4–1.7 μm · R ≈ 100–500 · 40 mas spaxels · 3 h

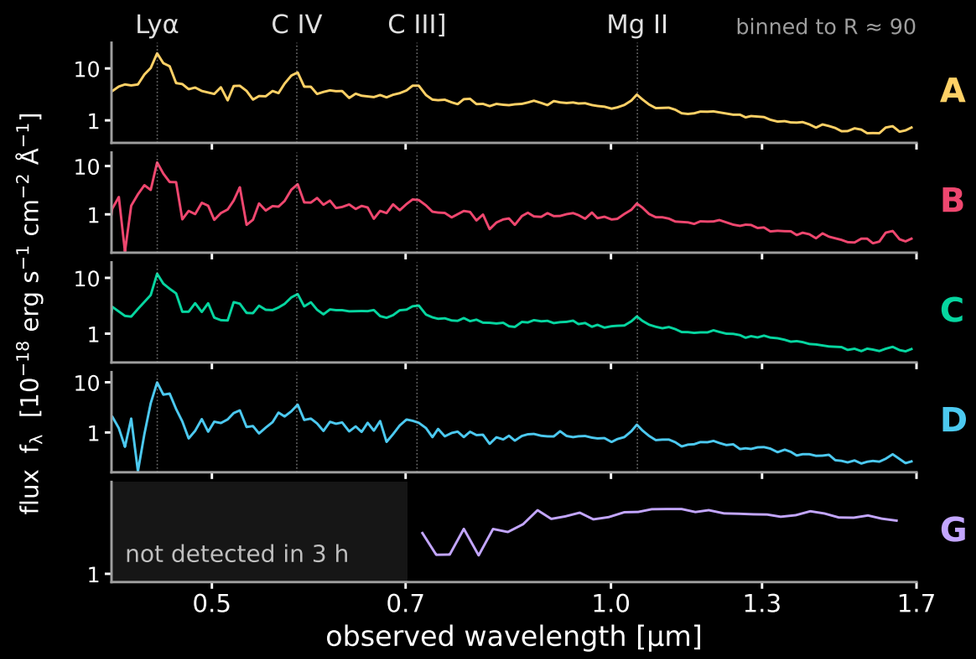


1.20–1.50 μm
lens galaxy: red starlight

0.85–1.00 μm
lens galaxy appears

0.60–0.70 μm
quasar images

Same quasar lines in every image → lensing confirmed



Simulated with SLSim; quasar hosts from HST COSMOS. Lensed-AGN counts: Oguri & Marshall 2010.
Lazuli: Roy et al. 2026 (arXiv:2601.02556). Modelled, not official: diffraction-limited 3.06 m Airy PSF + 4 mas jitter, SDSS-shaped filters, throughput, noise, exposure times.

Credits: Paras Sharma (Stony Brook University)